

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	M Vishnu Vardhan	Reg. No.:	192472087
Section / Batch:	2024	Date:	22/07/2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions	Marks
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – CONCEPT MAPPING

Q45. Consider the concept map: Recurrence Relation → Expansion → Pattern → Summation → Complexity Explain how recurrence expansion reveals patterns. Extend the map with substitution method link.

Justify importance of pattern recognition. Interpret how this supports solving recurrences.

Q46. Consider the concept map: Divide and Conquer → Subproblems → Recurrence → Complexity Map

how problem division leads to recurrence. Extend with parameters a and b. Justify their role in complexity.

Interpret how this connects design and analysis.

Code of Conduct

I M Vishnu Vardhan (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M Vishnu

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks (Each – 50)
Q1						
Q2						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CONCEPT MAP

M. S. Vardhan
19/04/2021

48)

Recurrence

Relation

→

Expansion

→

Pattern

→

Summation

Recurrence Relation

$$T(n) = 2T(n/2) + n$$



Expansion

$$T(n) = 2T(n/2) + n$$

$$= 4T(n/4) + 2n$$

$$= 8T(n/8) + 3n$$

$$= \dots \text{ (unrolls levels) }$$



SIMMATION

$$T(n) = n(1) + n \log n$$



Complexity

$$T(n) = O(n \log n)$$

Substitution Method Link

PATTERN ($n \log n$ form)



SUBSTITUTION

Step 1: Guess $T(n) = O(\log n)$

Step 2: Assume true for $n/2$

Step 3: Prove true for n

Pattern Recognition Matters

Pattern Recognition



Converts recursion $\rightarrow$ closed form

Links depth (k)

Gives correct guess

Reusable across similar recurrence

Expansion $\rightarrow$ Pattern $\rightarrow$ Guess $\rightarrow$ Proof $\rightarrow$ Verified

46) Divide and Conquer

Divide and Conquer
Break Problem size n



SUB Problems

'a' sub-problems
each of size n/b



Recurrence

$$T(n) = aT(n/b) + f(n)$$



Complexity

$$T(n) = O(n^a \log_b n)$$

Algorithm design $\rightarrow$ Analysis

$$T(n) = aT(n/b) + f(n)$$



Complexity Result



Feeds back $\rightarrow$ gives better design